Cup Sealing Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cup Sealer; Leg cup sealing machine

What it is used for

A cup sealing machine closes compatible cups with a heat-sealed film or lid to help prevent spills, contamination, and product leakage. Correct film selection, temperature and time settings, hygienic filling, and seal inspection help maintain packaging quality for food and beverages.

Main parts

- Machine frame supports the sealing assembly.
- Cup holder or mould positions the cup.
- Film roll or lid holder supplies sealing material.
- Heating plate or sealing head bonds the film to the cup rim.
- Temperature controller regulates sealing heat.
- Timer or cycle control sets sealing duration where fitted.
- Film feed and cutting mechanism advances and trims the film on applicable models.
- Control panel and switches operate the machine.
- Safety guard protects against moving parts and hot surfaces where fitted.

Before you start

1. Only trained personnel should operate the cup sealing machine.
2. Read the manual and confirm compatibility between cup material, rim shape, film, and product.
3. Inspect the frame, sealing head, heater, film feed, cutter, guards, and controls for damage.
4. Clean and sanitise food-contact surfaces according to the approved hygiene procedure.
5. Check that cups are undamaged, correctly filled, and free of product residue around the rim.
6. Verify the recommended sealing temperature, time, pressure, and film alignment for the packaging materials.
7. Wear suitable heat-resistant PPE where required and keep hands clear of the hot sealing head.

How to use

1. Place the machine on a stable, dry surface and connect power according to the manual.
2. Install the compatible film roll or lid and thread it through the feed mechanism as instructed.
3. Set the sealing temperature, time, and other permitted parameters according to the validated packaging procedure.
4. Allow the heating plate to reach its specified operating temperature where applicable.
5. Fill the cup according to the product process and keep the sealing rim clean and dry.
6. Position the cup correctly in the holder or mould and align the film over the rim.
7. Activate the sealing cycle and keep hands away from the heater, cutter, and moving components.

8. Allow the cycle to finish before removing the sealed cup from the holder.
9. Inspect the seal for wrinkles, incomplete bonding, misalignment, film damage, and visible leakage.
10. Perform the specified seal-integrity checks on representative cups and record the results.
11. Stop the machine and isolate power before clearing film jams, cleaning the sealing head, or servicing components.
12. Clean and sanitise the required surfaces after production and store unused packaging material in a clean, protected area.

⚠ Safety

- Keep fingers away from the hot sealing plate, cutter, and moving film-feed parts.
- Do not operate the machine with damaged wiring, faulty controls, or missing guards.
- Never reach into the machine to clear a jam while it is powered.
- Use only cups and films rated for the product and sealing process.
- Avoid overheating packaging materials and follow manufacturer limits for temperature and cycle time.
- Maintain food hygiene and prevent contamination of cup rims, film, and finished products.

Care & storage

- Switch off, isolate power, and allow hot parts to cool before cleaning.
- Remove film or product residue using manufacturer-approved methods that do not damage the sealing surface.
- Inspect the heater, cutter, film rollers, cup holder, guards, and cables regularly.
- Check temperature controls and sealing performance at the prescribed intervals.
- Arrange authorised maintenance if sealing becomes inconsistent or the heater fails to regulate temperature.

Common mistakes to avoid

- Using film or cups that are incompatible with the sealing process.
- Allowing liquid or food residue to remain on the cup rim.
- Setting incorrect temperature, time, or pressure and producing weak seals.
- Removing a cup before the cycle finishes or cleaning while the heater is hot.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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